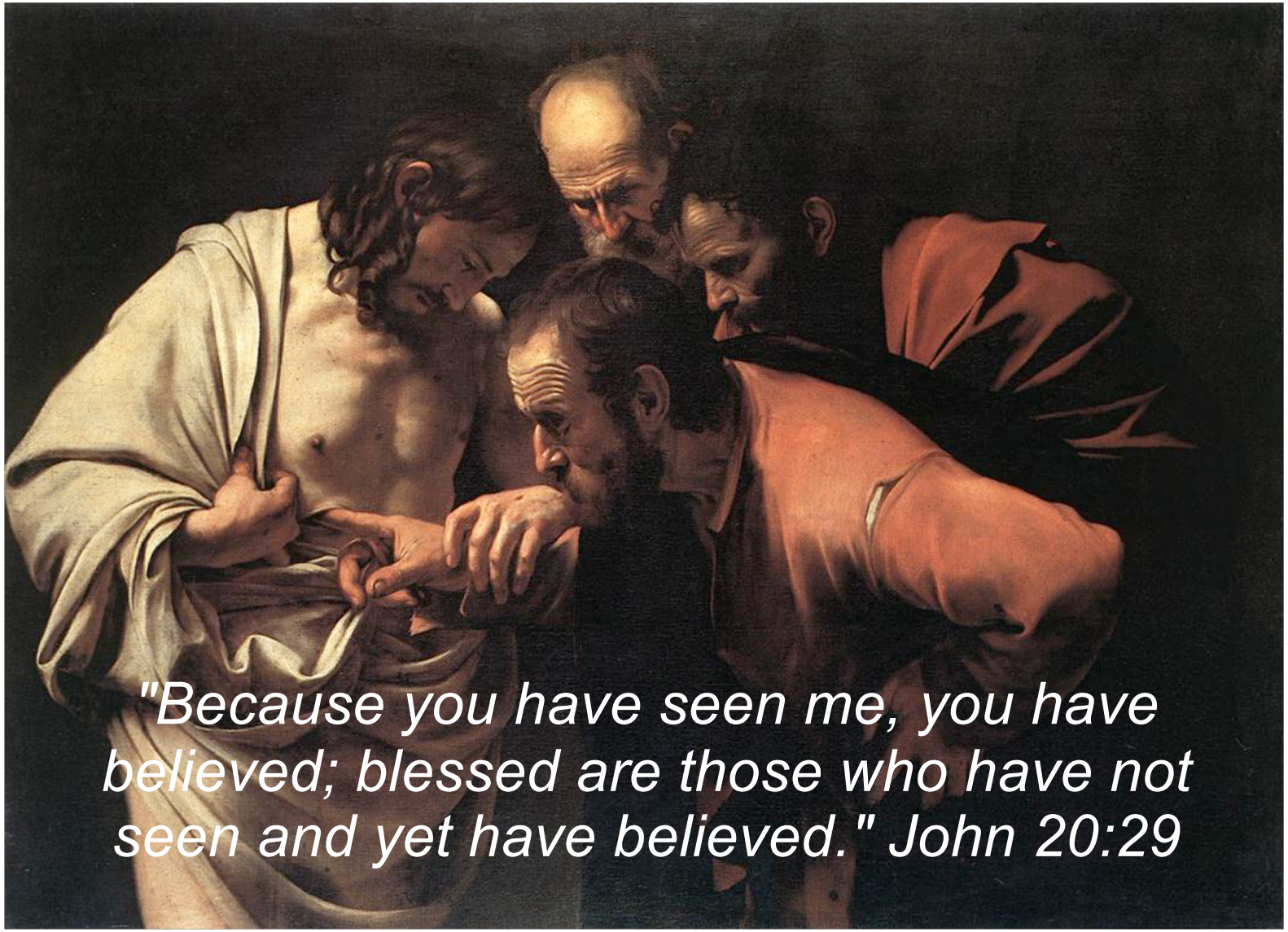
Observation and Measurement

Till Grüne-Yanoff

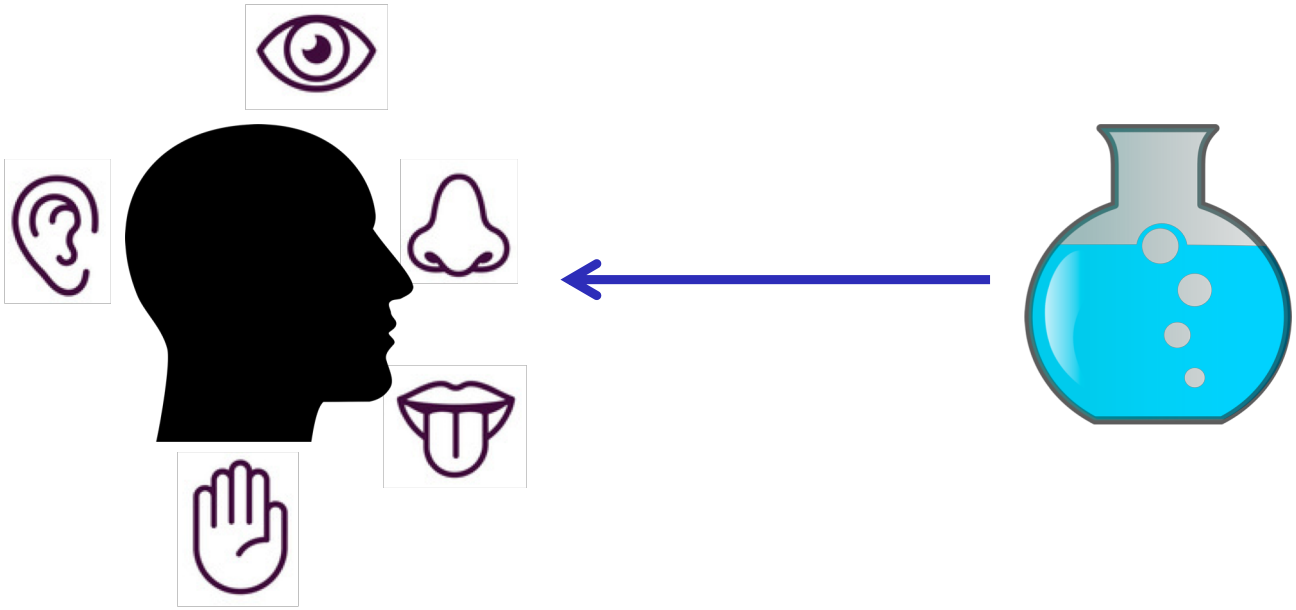
Empiricism and the Source of Scientific Knowledge

ἐμπειρία (*empeiria)* = sensory experience



Michelangelo Merisi da Caravaggio (1601) *Incredulità di San Tommaso*

Direct observation: by unaided sense experience

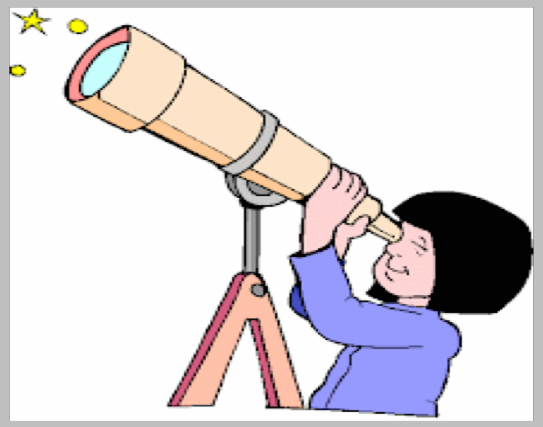


Direct observation: by unaided sense experience

Aided direct observation: by aided sense experience

Examples of aided direct observation:

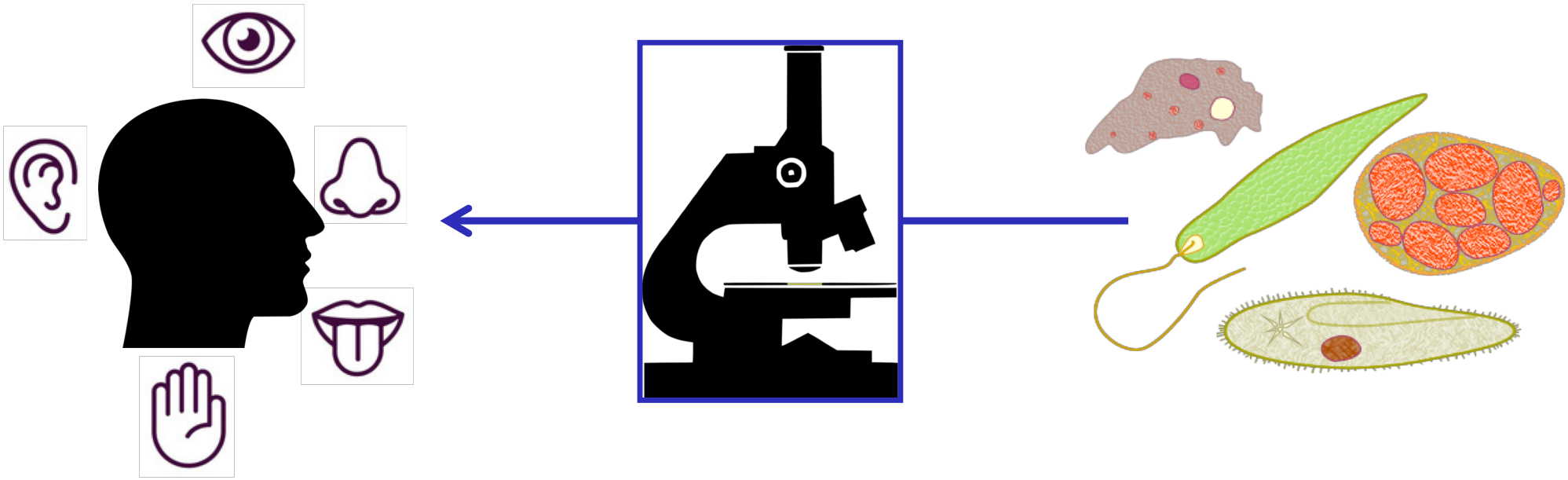
Optical microscope



Optical telescope

Direct observation: by unaided sense experience

Aided direct observation: by aided sense experience

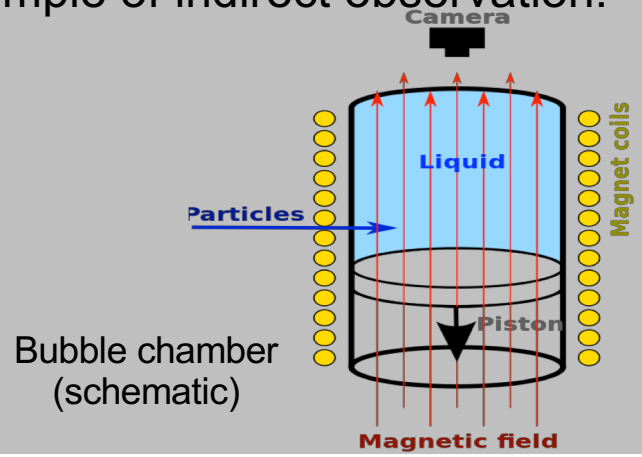


Direct observation: by unaided sense experience

Aided direct observation: by aided sense experience

Indirect observation: by (un)aided sense experience of the effect of an event, but not the event itself

Example of indirect observation:



Bubble chamber (schematic)

The first tracks observed in John Wood's liquid hydrogen bubble chamber, in 1954.

Direct observation: by unaided sense experience

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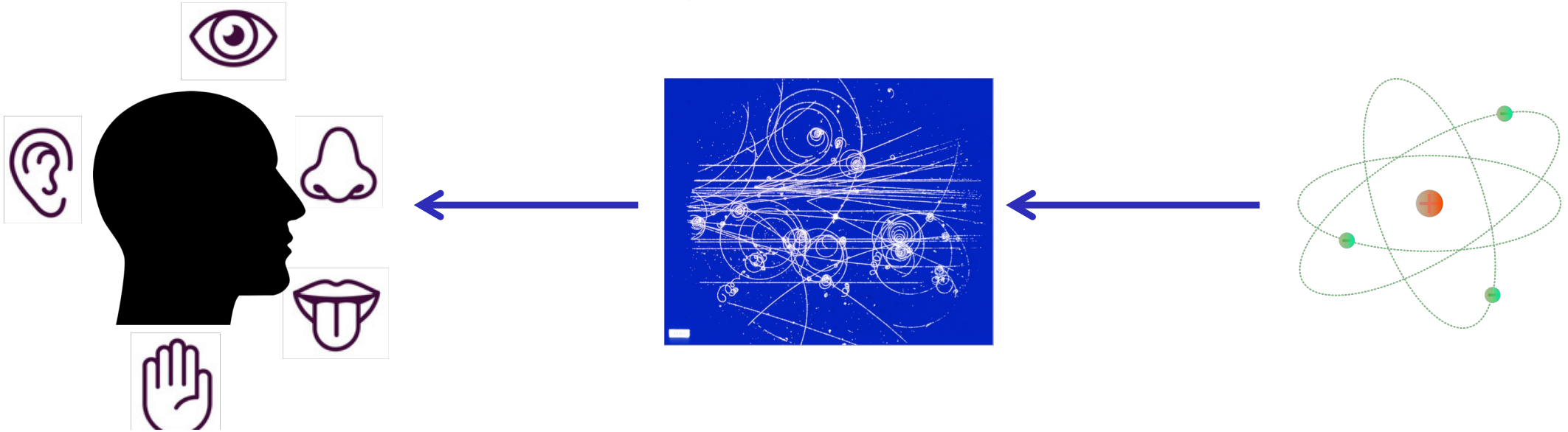


Summary

- Different kinds of sensory experience
- Different relations between observed object and senses
- Instruments play different roles

CONCEPTEST 1

Which of the following are *only indirectly* observable?

A. Comparative speed of two cars

B. Comparative speed of impulses traveling through a motor nerve

C. Bacteria

D. Temperature of a liquid

E. Happiness of a person

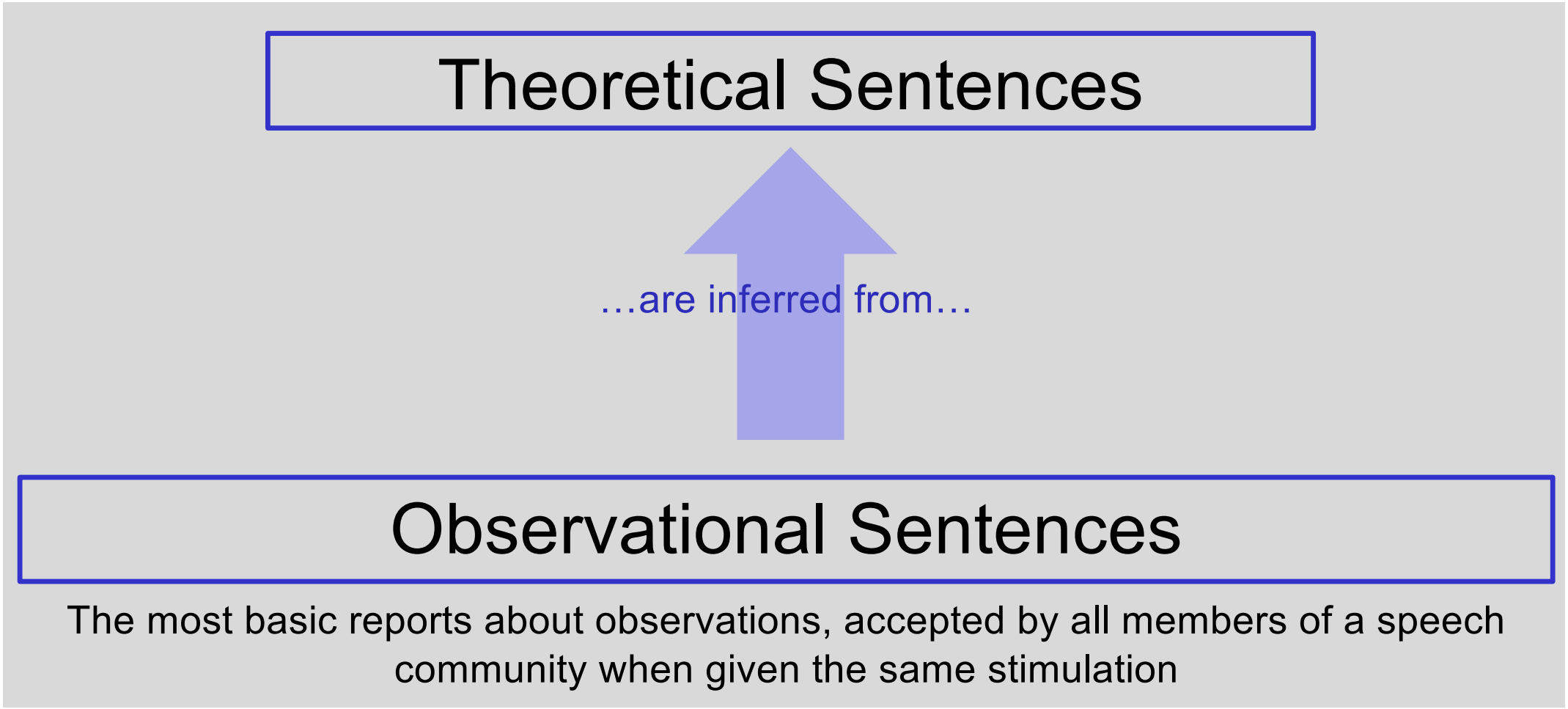
F. Stars of the Milky Way

G. Acidity of a liquid

Time allowed:
1’30

The Theory-Dependence of Observation

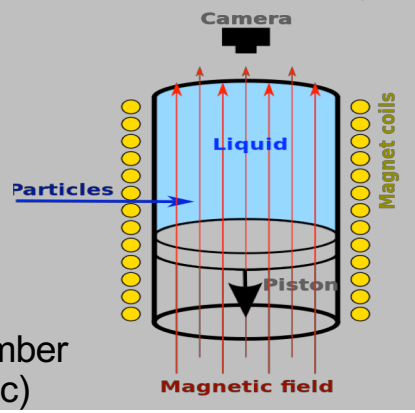
Logical Empiricism (1920s – 1950s)



The most basic reports about observations, accepted by all members of a speech community when given the same stimulation

Indirect Observation Requires Theory

Bubble chamber example again:



Bubble chamber (schematic)

Some required background assumptions:

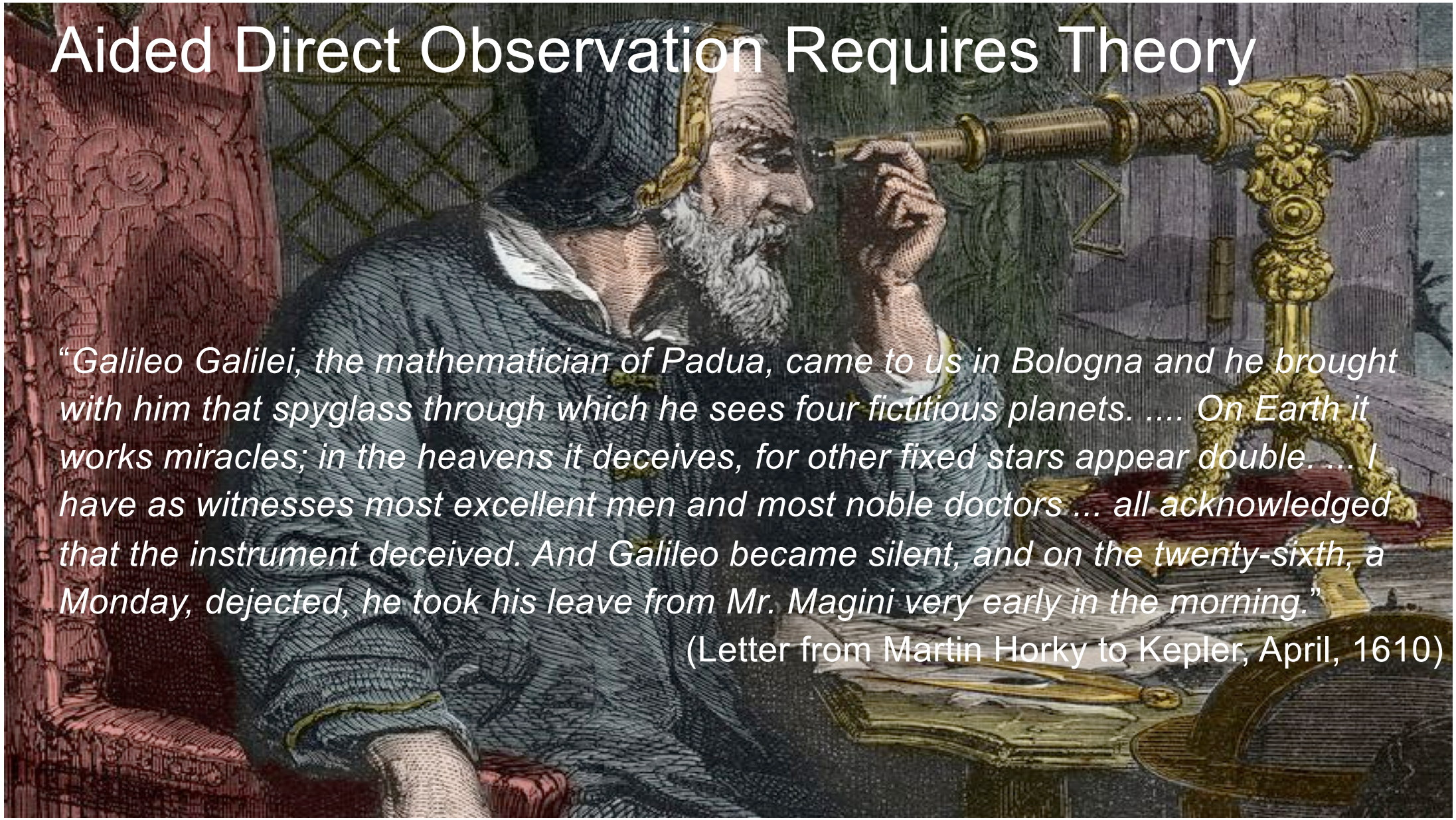
- Properties of liquids in superheated phase
- Relation particle charge – ionization
- Proportionality of bubble density to particle's energy loss.

Indirect observation: event of interest –(causes)→ observed effect

In order to associate event of interest with observed effect, we need theoretical knowledge

=>Indirect observation always theory-dependent

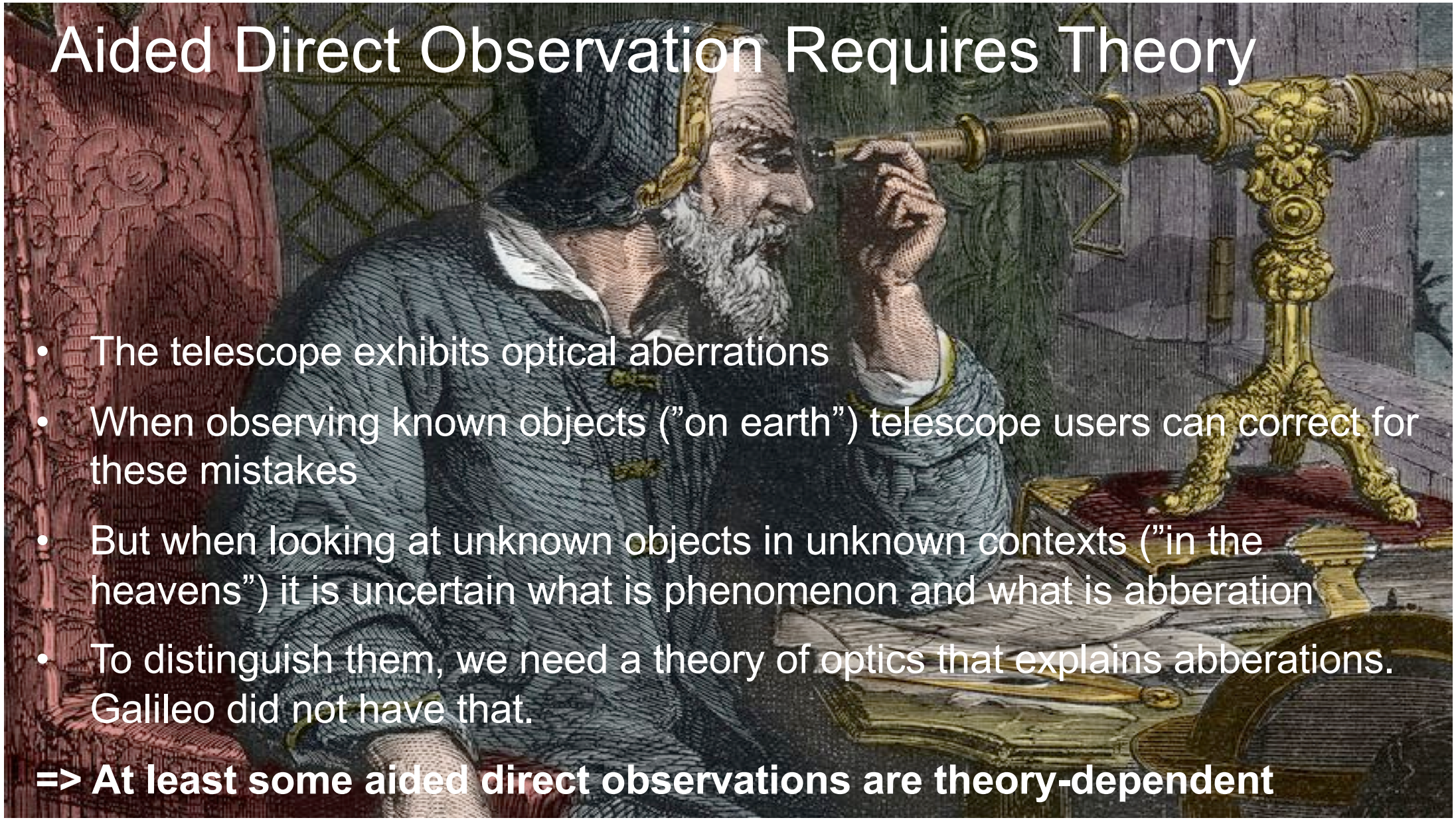
Aided Direct Observation Requires Theory



"Galileo Galilei, the mathematician of Padua, came to us in Bologna and he brought with him that spyglass through which he sees four fictitious planets. .... On Earth it works miracles; in the heavens it deceives, for other fixed stars appear double. ... I have as witnesses most excellent men and most noble doctors ... all acknowledged that the instrument deceived. And Galileo became silent, and on the twenty-sixth, a Monday, dejected, he took his leave from Mr. Magini very early in the morning."

(Letter from Martin Horky to Kepler, April, 1610)

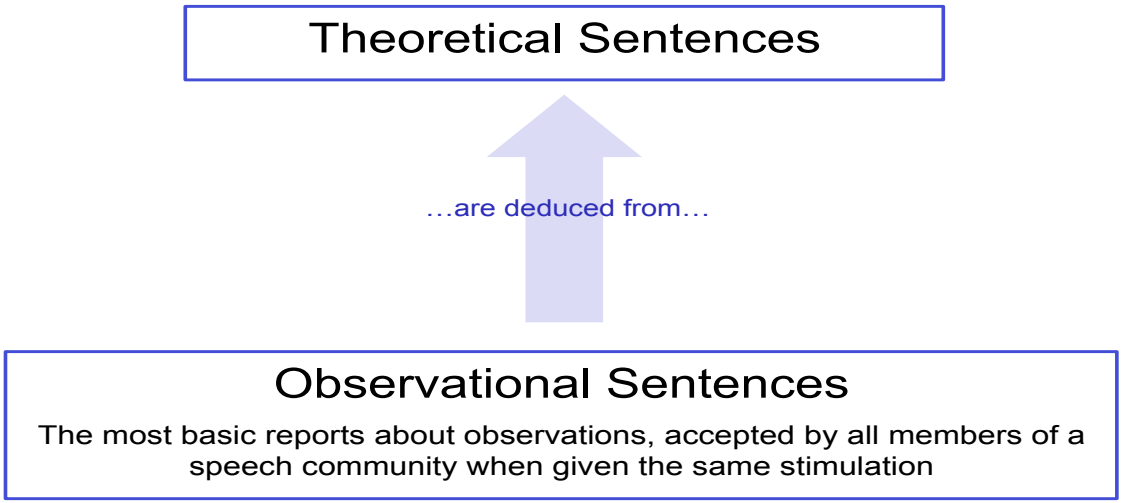
Aided Direct Observation Requires Theory



- The telescope exhibits optical aberrations
- When observing known objects (”on earth”) telescope users can correct for these mistakes
- But when looking at unknown objects in unknown contexts (”in the heavens”) it is uncertain what is phenomenon and what is abberation
- To distinguish them, we need a theory of optics that explains abberations. Galileo did not have that.

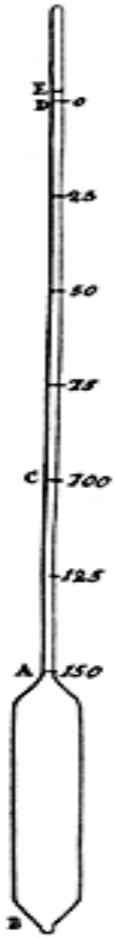
=> At least some aided direct observations are theory-dependent

If Observation is theory-dependent, then something is wrong with this picture:



…because theoretical claims were already used to generate observations

Circularity of Indirect Observation?



Example: temperature measurement

- Temperature –(causes)→ Density of liquid
- Assume: linear relationship
- What evidence do we have this relationship? Temperature measurements!
- But thermometers filled with different liquids did not agree

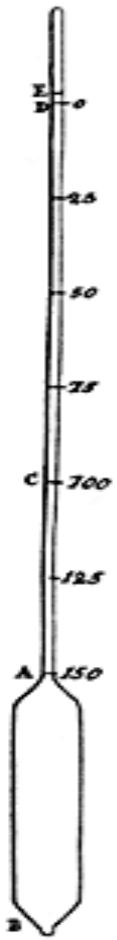
Mercury	Alcohol	Water
0 (°C)	0	0
25	22	5
50	44	26
75	70	57
100	100	100

(Chang, pg. 58.)

Circularity: Problem of Nomic Measurement

1. We want to measure quantity X.
2. Quantity X is not directly observable, so we infer it from another quantity Y, which is directly observable.
3. For this inference we need an empirical regularity that expresses X as a function of Y: X=*f*(Y).
4. The form of this function *f* cannot be discovered or tested empirically, because that would involve knowing the values of both Y and X, and X is the unknown variable that we are trying to measure.

Avoiding Circularity?



Henri Victor Regnault (1810-1878): Comparability Criterion

- Compare stability of observations from different thermometers:
  - Mercury thermometer fails:

Air thermometer	Mercury with "Choisy-le-Roi" crystal	Mercury with ordinary glass	Mercury with green glass	Mercury with Swedish glass
100 (°C)	100.00	100.00	100.00	100.00
150	150.40	149.80	150.30	150.15
200	201.25	199.70	200.80	200.50
250	253.00	250.05	251.85	251.44
300	305.72	301.08	--	--
350	360.50	354.00	--	--

(Chang, pg. 80.)

  - Air thermometer meets comparability:

Air thermometer A		*Air thermometer A'*		
Pressure (mmHg)	Temperature reading (°C)	Pressure (mmHg)	Temperature reading (°C)	Temperture difference (*A* - *A'*)
762.75	0	583.07	0	0
1027.01	95.57	782.21	95.57	0.00
1192.91	155.99	911.78	155.82	+0.17
1346.99	212.25	1030.48	212.27	-0.02
1421.77	239.17	1086.76	239.21	-0.04
1534.17	281.07	1173.29	280.85	-0.22
1696.86	339.68	1296.72	339.39	-0.29

(Chang, pg. 81.)

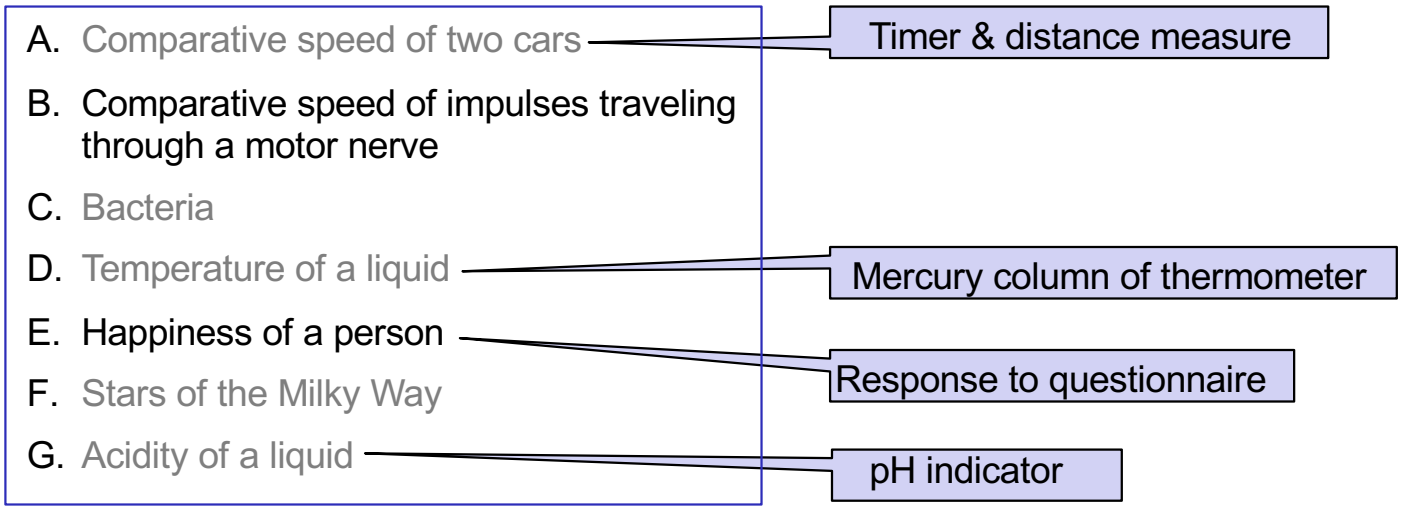
  - Conclude: Rely on air thermometer

Summary

- Indirect observation always theory-dependent
- Aided direct observation sometimes theory-dependent
- Theory-dependence complicates empiricism
- Threat of circularity of indirect observation

Operationalization

Only a few of the examples we discussed earlier were not directly observable in principle:



But in practice, scientists often use measurement instruments that make effects observable in a precise way – and these instruments typically provide indirect observations.

- Direct observation is prone to error, e.g.
  - Length perception varies with distance
  - Loudness perception varies with background noise
- Direct observation typically gives only qualitative properties

To *operationalize* a property of interest is to provide a way of linking it to a directly observable effect.

Infer: **property of interest**	Hypothesize: **–(causes)→**	Directly Observe: **Effect**
Temperature	*Linear relation of mercury density and temperature*	*Mercury column height*
Attitudes	*Causal relation between attitudes and answers*	*Questionnaire answers*

To *operationalize* a property of interest is to provide a way of linking it to a directly observable effect.

Examples of bad operationalizations

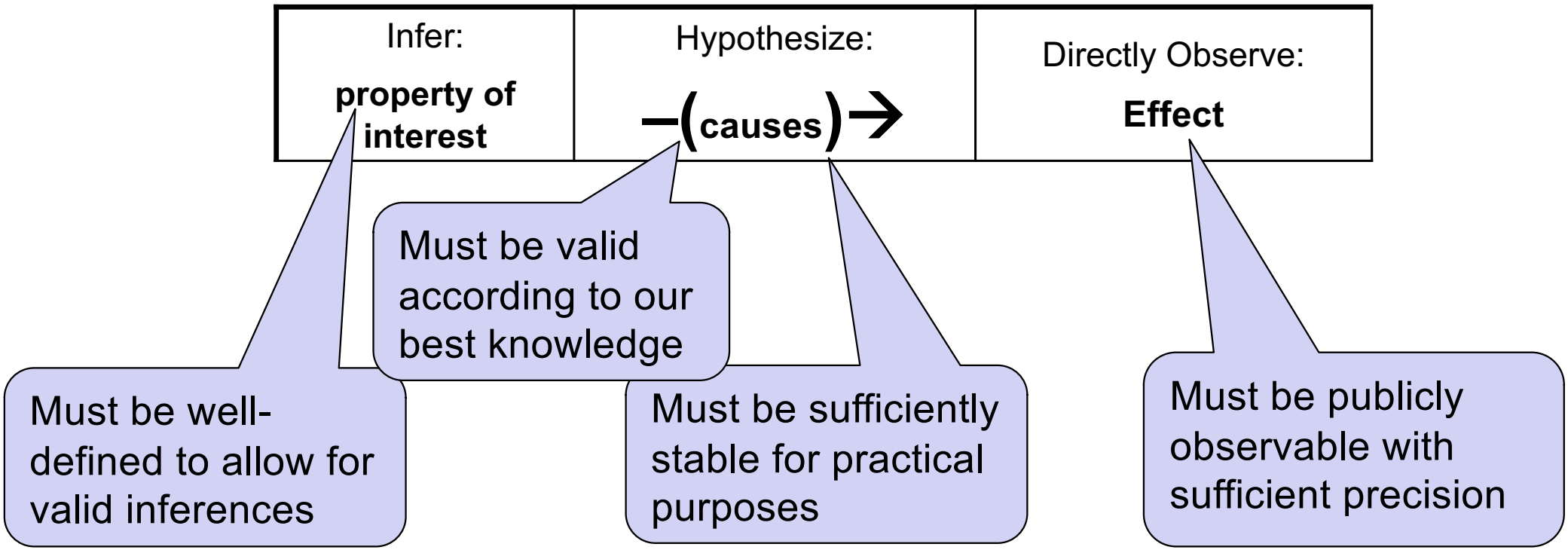
Infer: **property of interest**	Hypothesize: **–(causes)→**	Directly Observe: **Effect**
Mass of an object	*Mass of an object is directly proportional to the gravitational force exerted on that object*	

To *operationalize* a property of interest is to provide a way of linking it to a directly observable effect.

Examples of bad operationalizations

Infer: **property of interest**	Hypothesize: **–(causes)→**	Directly Observe: **Effect**
A person's voting decision	*People truthfully express their voting decisions in public*	*Expression of voting decision to a pollster*

Conditions for good operationalizations



Operationalism vs. Good Operationalizations

df: *Operationalism* is the view that even the most basic concepts are defined through the operations by which we observe/measure them.

Percy Williams Bridgman (1927)

Some arguments against *Operationalism*:

- We understand (empirical) concepts although for some, measurement techniques are enormously complex (e.g. very small lengths) or impractical (large distances like light years)
- Often want to criticize a measurement for not capturing the intended concept

Summary

- Indirect observation require operationalization
- Scientists often prefer to operationalize directly observable properties – i.e. observing them indirectly
- Quality criteria for operationalizations
- *Operationalism* does not help crafting good operationalizations

CONCEPTEST 2

What are necessary conditions for a good operationalization?

A. That it connects the property of interest with an observable property

B. That the concept to be measured is defined through the measurement operation

C. That the connection between property of interest and observable property is valid according to our best knowledge

D. That it is only applied to properties directly observable with aids

E. That the connection between property of interest and observable property is sufficiently stable for practical purposes

F. That the observable property is observable with sufficient precision in practice

Time allowed:
1’30

Measurement

What is Measurement?

i. Definition of concept to be measured ✔

ii. Specification of measurement operation(s) ✔

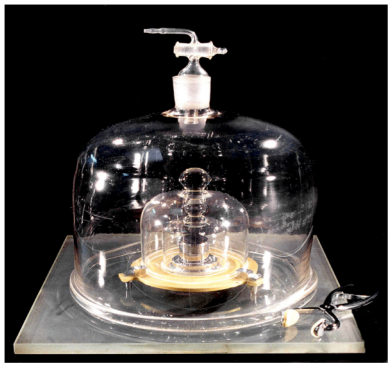
iii. Specification of a measure and definition of a unit

iv. Representation by numbers

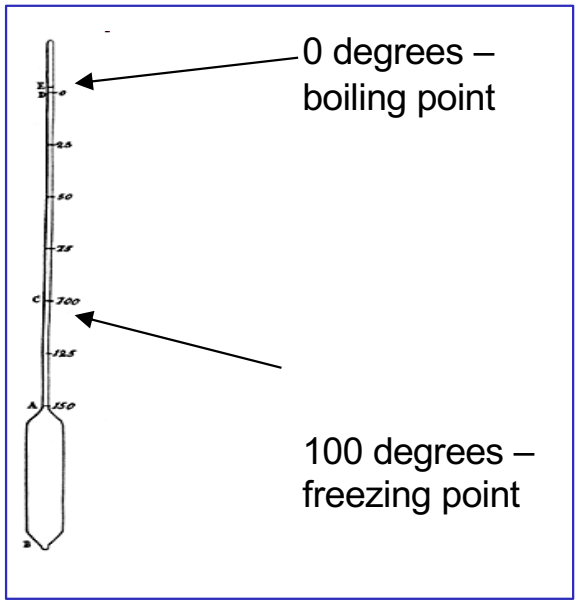
Measures

In order to represent the results of measurement operation with numbers, we need a comparison that provides a *unit*.

Legend for the first *metre* measurement, Vaugirard, Paris



Le Grand K - the international prototype of the kilogram. Forged in 1879 and sanctioned by the General Conference on Weights and Measures at its first meeting, kept in Paris.



Anders Celsius original thermometer

Requirements for Good Measures

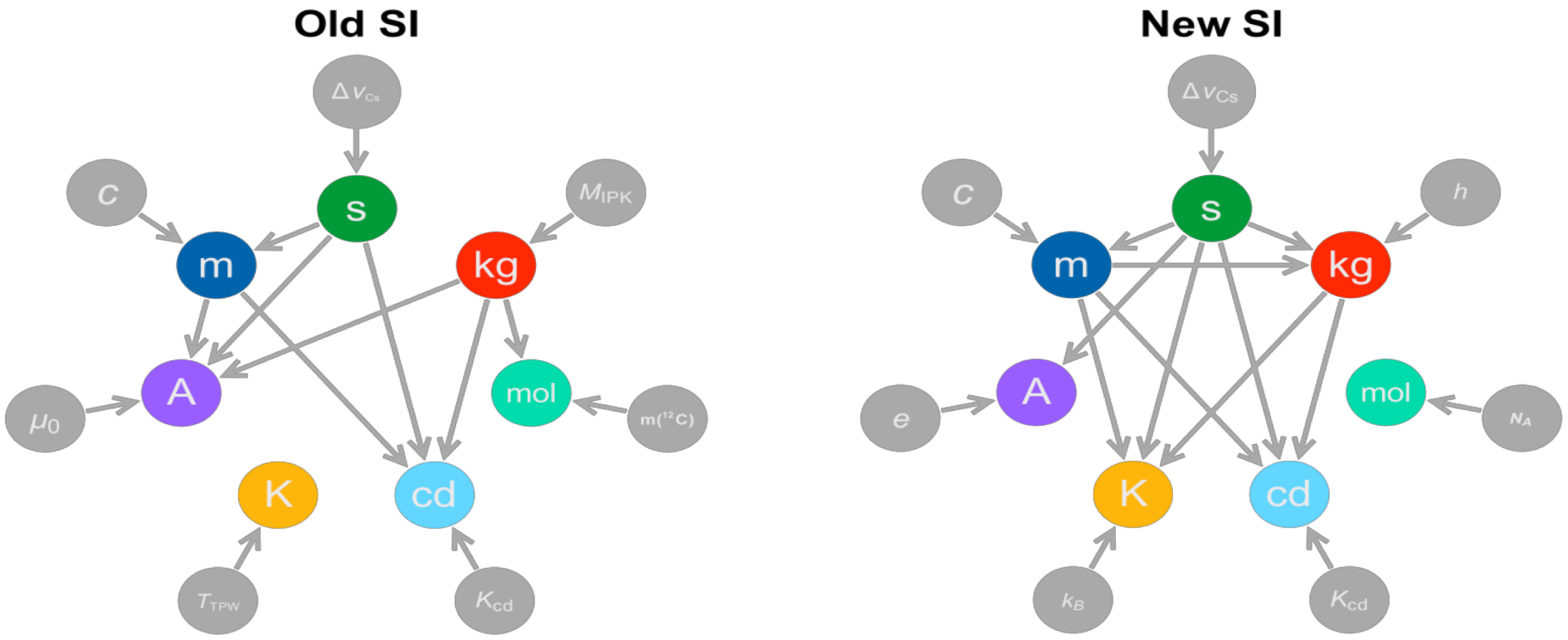
- The comparison unit must realize the relevant property (e.g. extension, mass, etc.)
- How much of the property is in a unit is set conventionally
- The measure must be sufficiently stable over time
- Everybody should be able to have access to the same measure

Definitions of the meter since 1795

Basis of definition	Date	Absolute uncertainty	Relative uncertainty
$^{1}/_{10,000,000}$ part of one half of a meridian, measurement by Delambre and Méchain	1795	0.5–0.1 mm	10^{-4}
First prototype *Mètre des Archives* platinum bar standard	1799	0.05–0.01 mm	10^{-5}
Platinum-iridium bar at melting point of ice (1st CGPM)	1889	0.2–0.1 μm	10^{-7}
Platinum-iridium bar at melting point of ice, atmospheric pressure, supported by two rollers (7th CGPM)	1927	n.a.	n.a.
1,650,763.73 wavelengths of light from a specified transition in krypton-86 (11th CGPM)	1960	0.01–0.005 μm	10^{-8}
Length of the path travelled by light in a vacuum in $^{1}/_{299,792,458}$ of a second (17th CGPM)	1983	0.1 nm	10^{-10}

International System of Units (SI)

Since 2018, all seven SI base units (in color) were defined by reference to natural constants. Until November last year, the kg was still defined by reference to an artifact - the international prototype.



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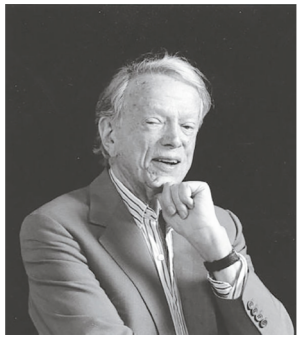
Representing with Numbers

"We cannot literally take a number in our hands and apply it to a physical object. What we can do is to show that the structure of a set of phenomena under certain empirical operations is the same as the structure of some set of numbers under arithmetical operations and relations"



Suppes 1967

Different scales types

Nominal scale

e.g. assigning samples identifying numbers

Ordinal scale

e.g. Mohs scale for hardness

Interval scale

e.g. temperature in Celsius

Ratio/quotient scale

e.g. length, volume, temperature in Kelvin

Absolute scale

e.g. counting objects, probability

What characterizes a scale type?

- Different scales of the same type are mathematically distinct, but represent identical empirical structures

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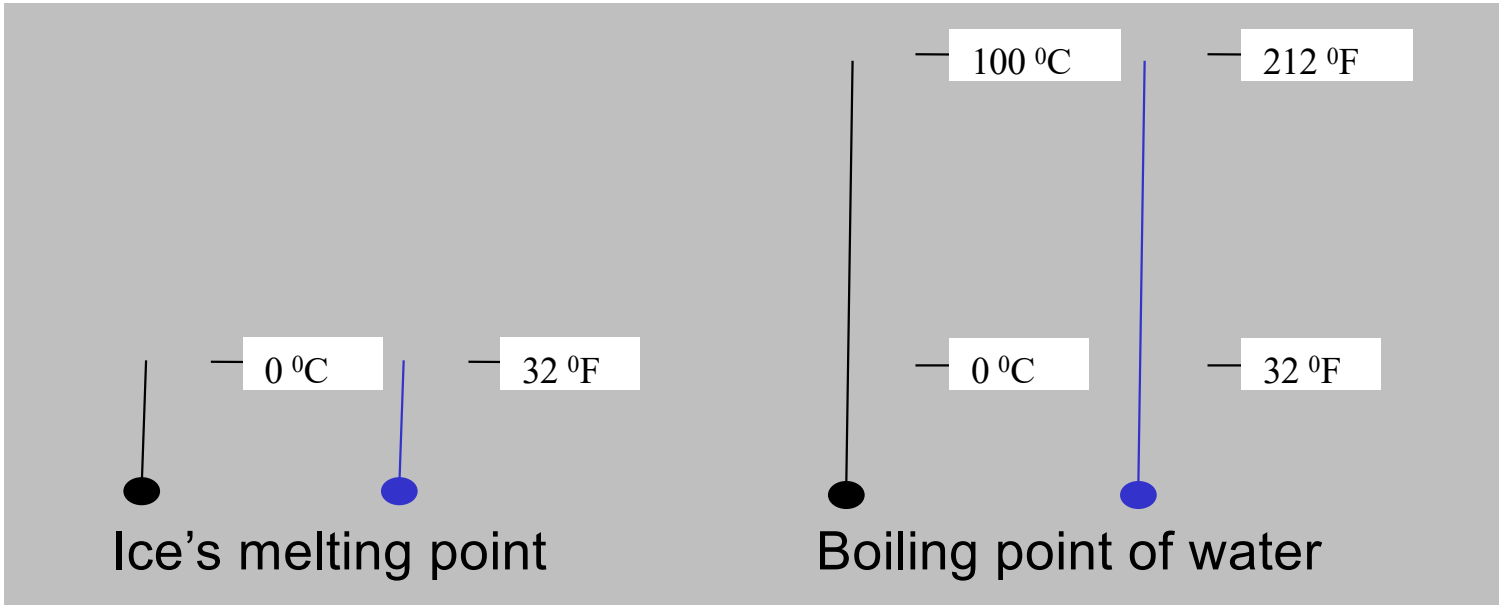

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- i.e. we can mathematically transform a scale without affecting the properties it represents

A measurement series with different scales

Celsius	⟷	*Fahrenheit*
5° C		41° F
10° C		50° F
15° C		59° F

Transformation: T(x° C) = (1.8*x + 32) ° F

"identical up to positive linear transformation"

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5°C x 2 = 10°C

However: *T*(5°C) x 2 = (41°F) x 2 = 82°F not the same as *T*(10°C) = 50°F

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Why meaningless?

Both F and C have an arbitrary zero point. To make claims about ratios of temperatures, we need a measure with an absolute zero.

=> Kelvin: 10K is indeed twice as warm as 5K.

What characterizes a scale type?

- Different scales of the same type are mathematically distinct, but represent identical empirical structures
- Example: *Celsius* (°C) and *Fahrenheit* (°F)
- i.e. we can mathematically transform a scale without affecting the properties it represents
- Which mathematical transformations are permitted determines the **type of scale**

Scales are distinguished by the kind of transformations they allow

- Nominal scale: **any that retains non-identity**
- Ordinal scale: **positive monotone transformation**
- Interval scale: **positive linear transformation**
- Ratio/quotient scale: **positive scalar transformation**
- Absolute scale: **unique**

Summary

- Requirements on Measures
- Scale types
- Kinds of transformations identifying scale types

CONCEPTEST 3

Identify the scale type for each of the following measurement examples!

1. Measuring Temperature in Kelvin
2. Assigning identifying numbers to different experimental batches
3. Recording the size of garments in S,M,L
4. Date measured in *AD*
5. Counting the number of people in a room

A. Nominal scale

B. Ordinal scale

C. Interval scale

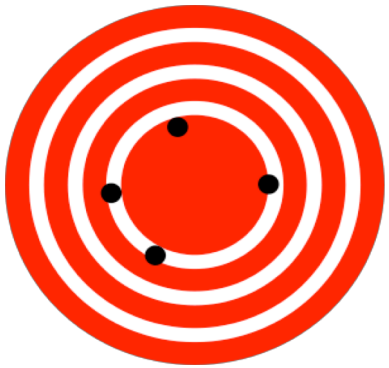
D. Ratio scale

E. Absolute scale

Time allowed:
1’30

Measurement Error

The difference between a measured value of a quantity and its true value



Two Dimensions of Error

Error type	Explanation		Contrasting quality
Systematic error (aka bias, inaccuracy)	Due to factors that should but are not controlled. Always occurs, with the same value, when applying the instrument in the same way to the same case.	Precise but inaccurate	Accuracy, validity
Random error (aka variation, imprecision)	Fluctuations in measurement results due to factors which cannot or will not be controlled. Have null expected value - i.e. they are scattered around the true value.	Accurate but imprecise	Precision, reliability

Dealing with Random Error

Random errors are inherently unpredictable, and have null expected value- i.e. they are scattered about the true value.

- They can be reduced by ***repeating measurements and averaging over outcomes***, or by trying an alternative operationalization that might have lower random errors.
- Reducing random errors increases precision, but not accuracy.

Dealing with Systematic Error

Systematic errors are predictable and typically constant or proportional to the true value.

- If the cause of the systematic error can be identified, then it usually can be eliminated.
- Reducing systematic errors increases accuracy, but not precision.

Identifying Systematic Error

Comparison between conceptual understanding and measurement result

- *Convergent validity*: Different ways of measuring the same intended concept should give rise to similar results.
- *Divergent validity*: Different ways of measuring different intended concepts should give rise to dissimilar results.

Avoiding Systematic Error

Control for potentially disturbing factors

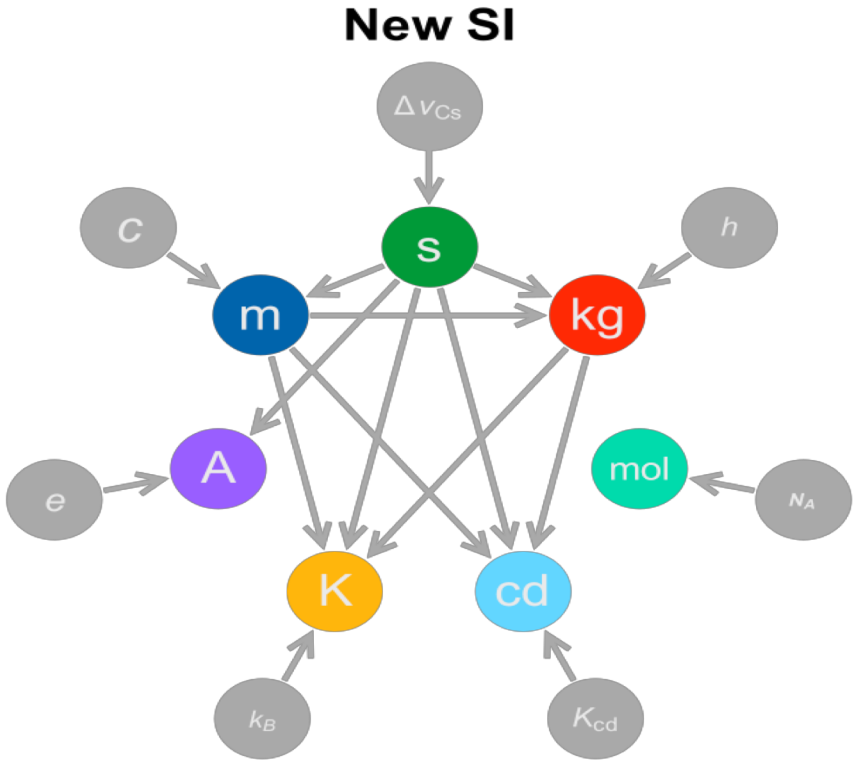
Example: observer influence

- A feature of interest can only be observed through its causal interaction with its environment
- To make a feature of interest observable is to change it.
- Goal: make causal effect of measurement procedure on measured object *negligible*

Avoiding Systematic Error

Standardize & stabilize measures

- Avoid unit incompatibilities
- Avoid unit deterioration



Measurement and Auxilliary Hypotheses

1. Formulate a hypothesis H

2. Deduce observable consequences $\{C_i\}$ from H, in conjunction with auxiliary hypotheses $\{AH_j\}$

$$H \;\&\; \{AH_j\} \;\rightarrow\; \{C_i\}$$

- Operationalisation ok
- No uncontrolled background factors
- Random error taken care of
- Unit stable
- Scale correct
- Instrument functioning

…

Summary

- Types of measurement errors
- Ways of identifying and avoiding errors

Learning Outcomes

- Distinguish direct and indirect observation
- Explain the concept of operationalization and apply its quality criteria
- Discuss the theory-dependence of observation and their consequences for logical empiricism
- Discuss the notion of a natural measure and the problem of nomic measurement
- Distinguish different measurement scales
- Distinguish systematic and random measurement error
- Understand the properties of accuracy and precision and apply strategies that improve measurements in these dimensions